Fishbone (Ishikawa) Diagram

Brainstorm every possible cause of a problem in six groups, so nothing is missed before you start testing.

REFERENCE	PART NUMBER OR PROCESS	DATE	LED BY	TEAM	LINKED 8D OR NCR

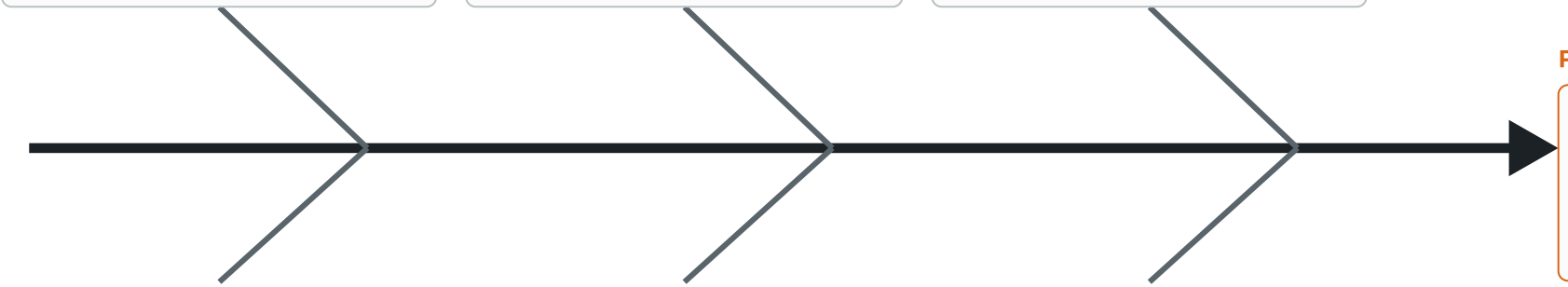
How to use this template

1. Write the problem in the orange box on the right, as a fact (what, where, how many).
2. With the team, list possible causes in each of the six boxes. Every idea goes down at this stage: no arguing yet.
3. Then pick the likeliest causes and test them, using the table on page 2.

People skills, training, fatigue

Machine equipment, tooling, software

Method procedures, work instructions



Problem (the effect)

Material components, PCBs, consumables

Measurement test, inspection, calibration

Environment temperature, humidity, ESD

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Next Pick the likeliest causes and test them

Don't fix anything until a cause is proven

- Circle the causes the team thinks most likely, using data where you have it.
- Test each one: change it and see if the problem follows.
- Take proven causes into 5 Whys or your 8D.

LIKELY CAUSE	HOW WE'LL TEST IT	OWNER	RESULT

Too many possible causes and no clear answer?

Grindstone Consultancy runs root cause workshops with your team, on site or remotely, at a fixed price agreed before we start.

enquiries@grindstone.ltd · grindstone.ltd/consultancy

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